ALGAMES: A Fast Solver for Constrained Dynamic Games

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Motivation

Dynamic games are an effective paradigm for dealing with the control of multiple interacting actors. Current algorithms either become intractable in high-dimensional real-world applications or cannot handle constraints or suffer from stability and tractability issues.
We introduce ALGAMES (Augmented Lagrangian GAME-theoretic Solver), a solver that handles trajectory optimization problems with multiple actors and general nonlinear state and input constraints. We present simulations demonstrating the speed and the ability of the solver to produce efficient, safe, and natural autonomous behaviors in the context of autonmous driving.

Contributions

- A general solver for dynamic games aimed at identifying generalized Nash equilibrium strategies.
- Demonstration of the solver's speed, robustness, and scalability in autonomous driving applications.

Game theory

Game with **M** players, each chooses a strategy p^ν that solves a minimization problem with constraints:

$$\min_{p^\nu} \quad J^\nu(p^\nu, \boldsymbol{p}^{-\nu}),$$
$$\text{s.t.} \quad C^\nu(p^\nu, \boldsymbol{p}^{-\nu}) \leq 0.$$

We form the augmented Lagrangian for each player:

$$L^\nu(p^\nu, \boldsymbol{p}^{-\nu}) = J^\nu + \lambda^{\nu T} C^\nu + \frac{1}{2} C^{\nu T} I_{\rho^\nu} C^\nu.$$

We express optimality of each player using the 1st order necessary condition:

$$\nabla_{p^\nu} L^\nu(p^\nu, \boldsymbol{p}^{-\nu}) = G^\nu(p^\nu, \boldsymbol{p}^{-\nu}) = 0.$$

We obtain a root–finding problem encoding the fact that all players act optimally:

$$\min_{\boldsymbol{p}} \quad 0,$$
$$\text{s.t.} \quad G^\nu(p^\nu, \boldsymbol{p}^{-\nu}) = 0, \ \forall\, \nu \in \{1, \ldots, M\}.$$

1. We solve it using a Gauss-Newton method:

$$\delta \boldsymbol{p} = -H^{-1} G. \qquad H = \nabla_{\boldsymbol{p}} G$$

2 .We update the Lagrange multipliers and the penalty terms:

$$\lambda_k^\nu \leftarrow \lambda_k^\nu + \rho_k^\nu C_k^\nu(\boldsymbol{p})$$
$$\rho_k^\nu \leftarrow \gamma \rho_k^\nu, \ \forall k \in \{1, \ldots, n_c\}.$$

3. We repeat steps 1 and 2 until convergence.

At convergence, each player's strategy is a best response to the other players' strategies, we get a generalized Nash equilibrium.

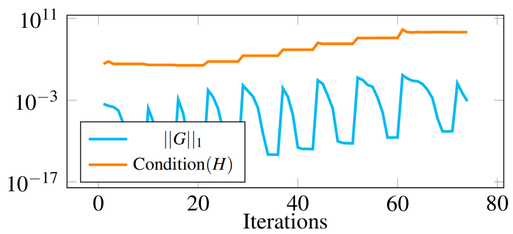
Paper: `https://rexlab.stanford.edu/papers/ALGAMES.pdf`
Code: `https://github.com/RoboticExplorationLab/ALGAMES.jl`

ALGAMES

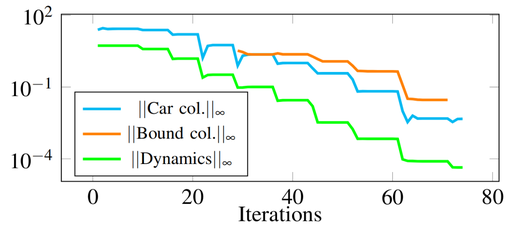
Algorithm 3 ALGAMES solver

procedure ALGAMES($\boldsymbol{p}_0, \rho_0$)
  Initialization
  $\lambda^\nu \leftarrow 0,$ $\quad \forall \nu$
  $\rho^\nu \leftarrow \rho_0,$ $\quad \forall \nu$
  $\boldsymbol{p} \leftarrow \boldsymbol{p}_0$
  Until Convergence **do**
    $\boldsymbol{p} \leftarrow$ NEWTON'SMETHOD($\boldsymbol{p}$)
    $\lambda^\nu \leftarrow$ DUALASCENT($\boldsymbol{p}, \lambda^\nu, \rho^\nu$), $\quad \forall \nu$
    $\rho^\nu \leftarrow$ INCREASINGSCHEDULE(ρ^ν), $\quad \forall \nu$
  return $\boldsymbol{p}$

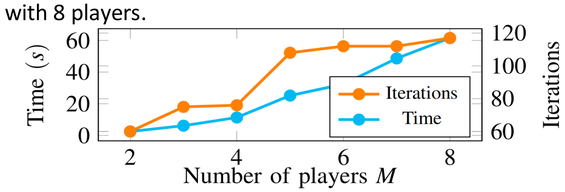
ALGAMES solves the root-finding problem robuslty: we observe that the condition number of H incrementally increases after each penalty update. However, it remains in a reasonable range during the solve. This reasonable conditioning of the numerical problem, ensures the robustness of the solver.



ALGAMES handles nonlinear constraints: we plot constraint satisfaction; we observe a linear convergence of the maximum constraint violations.

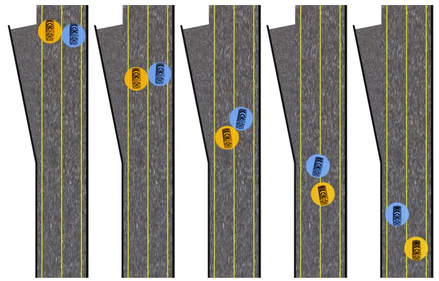


ALGAMES is highly scalable: the compute time remains reasonable even with 8 players.

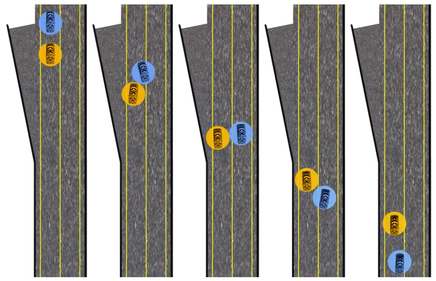


Experiments

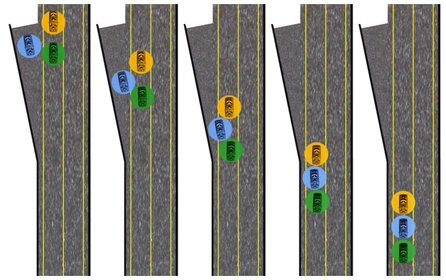
We tested ALGAMES in the autonomous driving context. Some situations like ramp merging involve a high level of interaction between the vehicles. In these situations, the game-theoretic nature of the problem cannot be ignored.
We used 'egoistic' objective functions for each player and added collision avoidance constraints with the other vehicles, as well as with the road boundaries.
We obtained natural-looking trajectories where the vehicles share the responsibility of avoiding collisions and demonstrate rich autonomous behavior.



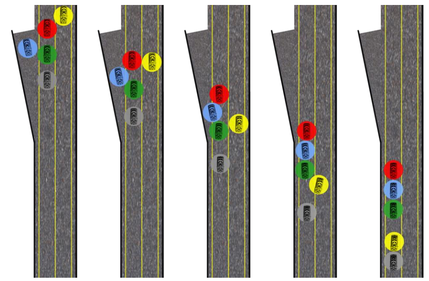
Lane change: the collision avoidance responsibility is shared.



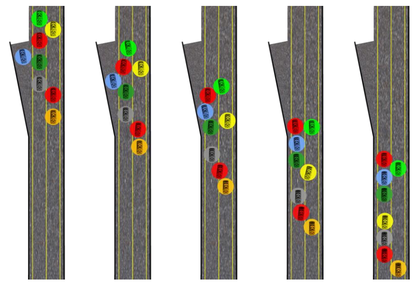
Overtaking: natural trajectories simply triggered by a difference in desired speed.



Ramp merging: blue vehicle squeezes between orange and green.



5 vehicles: 1 ramp merging, 1 lane change.



8 vehicles with high level of interaction.

Scenario	Time in s
Lane changing	2.3
Overtaking	4.2
Ramp merging	6.1
5 vehicles	25.2
8 vehicles	61.5

Future work:

- Parallelization of the code for real-time performance,
- Hardware implementation.